

Topology-Guided Model Predictive Path Integral Control for Dynamic Obstacle Avoidance on a Skid-Steer Robot

Saran Sundar

Abstract—Sampling-based model predictive control such as Model Predictive Path Integral (MPPI) control optimises a single distribution over control sequences. In environments containing moving obstacles this is a liability: the optimal manoeuvre may be to pass an obstacle on either side, and averaging samples drawn from both options yields a control that does neither. Topology-driven methods address this by optimising several trajectories in distinct homotopy classes in parallel. This report presents TG-MPPI, a topology-guided MPPI controller that constructs candidate guidance trajectories from a robot-centred geodesic flood field, allocates sample groups to them, and evaluates a separate free energy per group. TG-MPPI is implemented as a drop-in Nav2 controller plugin and evaluated against the unmodified Nav2 MPPI baseline on a simulated skid-steer robot in scripted dynamic-obstacle scenarios. **[TODO: one sentence of headline result once the re-run lands]** The evaluation isolates constant-velocity obstacle prediction from topological guidance through a controlled ablation. The final comparison is reported with a route-distinctness test based on relative winding, replacing an earlier test that was structurally unable to recognise pass-before/pass-behind alternatives.

Index Terms—Model predictive control, motion planning, dynamic obstacle avoidance, homotopy classes, mobile robots.

I. INTRODUCTION

SAMPLING-BASED predictive controllers have become the default local planner for many mobile robots operating in cluttered environments. They can optimise non-linear motion without requiring an analytic gradient and can evaluate thousands of candidate controls in parallel. Dynamic clutter remains difficult, however, because collision-free motion depends on both geometry and timing. A route that is safe now can be occupied when the robot arrives, while stopping, passing before an obstacle, and passing behind it may all be locally feasible but belong to qualitatively different strategies.

Standard MPPI maintains one nominal control sequence and updates it with a weighted average of sampled perturbations. This representation is effective when good samples form one coherent mode. Near an obstacle, however, samples may split between left/right or pass/wait alternatives. Their average need not represent either alternative and can point toward the obstacle or hesitate between routes. The dynamic scenes in Fig. 3 are designed to expose this interaction between spatial

congestion and obstacle timing on a rectangular skid-steer platform.

This work asks whether explicit local route modes improve MPPI beyond the benefit obtained from predicting obstacle motion alone. TG-MPPI constructs a robot-centred geodesic body from the local costmap, extracts locally distinct “pseudopod” branches, converts them to dynamically feasible ancillary control sequences, and assigns separate groups of MPPI samples to these proposals. A time-expanded search can additionally propose wait and detour strategies around predicted moving obstacles. The controller is implemented as a separate Nav2 plugin so that the stock Nav2 MPPI implementation remains an untouched baseline.

The contributions of this report are:

- 1) TG-MPPI, a topology-guided MPPI controller that derives candidate guidance trajectories from a geodesic flood field rather than from a separate global topological planner, implemented as a Nav2 plugin (Section IV);
- 2) a scripted, reproducible dynamic-obstacle benchmark with ground-truth contact scoring, in which every condition sees identical obstacle trajectories (Section VI);
- 3) an ablation separating the contribution of obstacle prediction from that of topological guidance (Section VII); and
- 4) an evaluation of space-time route guidance using a winding-based distinctness test, together with a diagnosis of why the earlier synchronised-time test was invalid (Section VII-C).

II. RELATED WORK

MPPI casts finite-horizon stochastic control as an importance-weighted update over sampled control perturbations and maps naturally to parallel hardware [1], [2]. Unlike local reactive methods such as the Dynamic Window Approach [3], it evaluates complete short trajectories rather than admissible instantaneous velocities. The standard formulation nevertheless represents its solution with one nominal sequence; it does not explicitly preserve multiple route classes when the rollout distribution becomes multimodal.

Biased-MPPI generalises the sampling step by allowing ancillary controllers to inject informative control sequences instead of drawing every candidate around the shifted solution from the previous cycle [4]. This can place samples near

braking, tracking or learned strategies that ordinary local Gaussian exploration might rarely discover. The ancillary controllers in that work are supplied by the designer or a learned policy. TG-MPPI builds on this idea but addresses a different question: how can the environment's currently reachable topology generate those ancillary strategies online? Its pseudopods and space-time routes answer that question, while grouped updates prevent opposed strategies from being averaged together.

Topology-aware planners instead reason about families of paths that cannot be continuously deformed into one another without crossing obstacles. H-signatures provide a formal algebraic representation for search-based planning [5], while distinctive-topology TEB optimises several elastic-band candidates online [6]. These methods motivate keeping alternatives separate until their quality can be compared, but they do not directly specify how a sampling-based controller should construct and allocate its rollout distribution.

The closest methodological reference is topology-driven parallel trajectory optimisation (T-MPC), which builds multiple candidate routes and optimises them in parallel in dynamic environments [7]. TG-MPPI adopts the same high-level principle—preserve and compare distinct decisions—but uses a rolling geodesic flood field rather than a probabilistic road map to create local spatial guidance. It then represents the alternatives as groups inside one MPPI rollout batch rather than as independent gradient-based trajectory optimisers. The result is packaged within Nav2 [8], making an unmodified Nav2 MPPI plugin the operational baseline rather than comparing against a separately reimplemented controller.

III. BACKGROUND AND PROBLEM FORMULATION

A. Information-theoretic stochastic control

Consider a robot with state $\mathbf{x}_t \in \mathbb{R}^n$ and control $\mathbf{u}_t \in \mathbb{R}^m$ evolving under $\mathbf{x}_{t+1} = f(\mathbf{x}_t, \mathbf{u}_t)$. For the horizon control sequence $U = (\mathbf{u}_0, \dots, \mathbf{u}_{T-1})$, define

$$S(U) = \phi(\mathbf{x}_T) + \sum_{t=0}^{T-1} \ell(\mathbf{x}_t, \mathbf{u}_t), \quad (1)$$

where S may include goal, path, constraint, static-obstacle and predicted dynamic-obstacle terms. Let P be a reference distribution over control sequences. The information-theoretic formulation seeks a distribution Q that minimises

$$\mathcal{J}(Q) = \mathbb{E}_Q[S(U)] + \lambda D_{\text{KL}}(Q \| P), \quad (2)$$

where $\lambda > 0$ trades expected trajectory cost against departure from P .

Introducing a multiplier for $\int q(U) dU = 1$ and setting the functional derivative of (2) to zero yields

$$0 = S(U) + \lambda \left[\log \frac{q^*(U)}{p(U)} + 1 \right] + \beta, \quad (3)$$

$$q^*(U) = \frac{1}{\eta} \exp \left[-\frac{S(U)}{\lambda} \right] p(U), \quad (4)$$

$$\eta = \mathbb{E}_P \left[\exp \left(-\frac{S(U)}{\lambda} \right) \right]. \quad (5)$$

Low-cost sequences therefore receive exponentially greater probability, while λ determines how sharply the distribution concentrates.

B. Classical MPPI

Classical MPPI samples N sequences $U^{(i)} = \bar{U} + \mathcal{E}^{(i)}$ around the shifted solution from the previous cycle and rolls each sequence through the dynamics. With the standard MPPI trajectory cost, including its control/noise term, the Monte-Carlo update is

$$w^{(i)} = \frac{\exp[-(S^{(i)} - S_{\min})/\lambda]}{\sum_{j=1}^N \exp[-(S^{(j)} - S_{\min})/\lambda]}, \quad (6)$$

$$\mathbf{u}_t^* = \bar{\mathbf{u}}_t + \sum_{i=1}^N w^{(i)} \epsilon_t^{(i)}. \quad (7)$$

Subtracting $S_{\min}$ is numerically stabilising and does not alter the normalised weights. The first updated command is applied, the remaining sequence is shifted forward, and the calculation repeats at the next cycle [1], [2].

C. Biased MPPI and ancillary controllers

The shifted previous solution is useful when the scene changes slowly, but it can concentrate the entire sample population around a newly invalid strategy. Biased-MPPI permits an alternative sampling distribution Q_s informed by ancillary controllers [4]. For samples $U^{(i)} \sim q_s$, the general importance-sampling form is

$$\tilde{w}^{(i)} = \exp \left[-\frac{S(U^{(i)})}{\lambda} \right] \frac{p(U^{(i)})}{q_s(U^{(i)})}, \quad (8)$$

$$w^{(i)} = \frac{\tilde{w}^{(i)}}{\sum_j \tilde{w}^{(j)}}, \quad \mathbf{u}_t^* \approx \sum_i w^{(i)} \mathbf{u}_t^{(i)}. \quad (9)$$

The density ratio corrects for drawing from Q_s rather than P ; equivalently, its logarithm can be absorbed into an augmented trajectory cost. For M ancillary controllers producing horizon-length means μ_m , the proposal can be represented as

$$q_s(U) = \sum_{m=0}^M \pi_m \mathcal{N}(U; \mu_m, \Sigma_m), \quad \sum_m \pi_m = 1, \quad (10)$$

where $m = 0$ is the nominal or global-path fallback. Biased-MPPI supplies the mechanism for injecting and fusing ancillary strategies, but it does not prescribe that their means must represent locally distinct collision-free routes. TG-MPPI's contribution begins there: it generates μ_m online from geodesic pseudopods and space-time alternatives.

For clarity, the sandbox includes the explicit density correction of (8), but the evaluated Nav2 controller uses the stable grouped V3 estimator described in Section V: conditional softmax weights inside each group followed by discrete group selection. It inherits Biased-MPPI's ancillary-proposal principle, but is not claimed to be an exact implementation of its fully importance-corrected estimator.

D. Multimodal failure addressed by TG-MPPI

Equation (7) still forms one control average. If similarly weighted samples pass an obstacle on opposite sides, their average can lie between the two modes and commit to neither. Let $\mathcal{O}_{0:T}$ denote the static and predicted dynamic obstacle set. Two feasible trajectories ξ_a and ξ_b are in the same space-time homotopy class when one can be continuously deformed into the other without intersecting $\mathcal{O}_{0:T}$ while their endpoints remain fixed. Partitioning samples into candidate classes $\mathcal{G}_g$ changes the decision into a conditional update followed by explicit selection,

$$g^* = \arg \min_g F_g, \quad \mathbf{u}_t^* = \sum_{i \in \mathcal{G}_{g^*}} w_{i|g^*} \mathbf{u}_t^{(i)}. \quad (11)$$

The objective is not to prove that every possible class has been enumerated, but to preserve a small set of useful local decisions long enough for MPPI to compare them without averaging opposed controls together.

IV. METHOD: TG-MPPI

A. Geodesic flood field and the robot-centred body

The local costmap is discretised at resolution Δr and obstacles are inflated by a circular planning-footprint radius r_p . The remaining free cells form an eight-connected graph, with axial edge cost Δr and diagonal cost $\sqrt{2}\Delta r$. A single-source Dijkstra expansion from the robot gives the collision-free geodesic distance

$$d_r(p) = \min_{\gamma: p_r \rightsquigarrow p} \sum_{e \in \gamma} w(e). \quad (12)$$

Dijkstra is used instead of A* because the distances to all reachable cells, not one known endpoint, are required. The geodesic body is the sublevel set

$$\mathcal{B}_\rho(p_r) = \{p \in \mathcal{F} : d_r(p) \leq \rho\}, \quad (13)$$

where ρ is a travel-distance budget rather than a Euclidean radius. It is approximately circular in open space, but deforms around inflated obstacles and pinches through connected passages. The body is rebuilt from the current robot pose at the configured field-update rate; it is therefore a rolling reachability construction, not a fixed shape translated through the map.

Its outward membrane contains body cells adjacent to reachable free space just beyond the budget,

$$\partial \mathcal{B}_\rho = \{p \in \mathcal{B}_\rho : \exists q \in \text{Adj}_8(p), q \in \mathcal{F} \setminus \mathcal{B}_\rho\}. \quad (14)$$

A wall-facing body edge is consequently not treated as a possible exit merely because it borders an obstacle.

Figure 1 shows three rebuilds produced by the actual Python sandbox in BARN world 48. The changing outline illustrates the key geodesic property: the body measures connected travel through free space and therefore changes shape rather than translating as a rigid Euclidean disc.

B. Pseudopod branches as guidance trajectories

Every membrane cell m receives a terminal cost-to-go, or *promise*, from the global path $\mathcal{P} = \{z_k\}$,

$$h(m) = \min_{z_k \in \mathcal{P}} (\|m - z_k\|_2 + L_{\text{rem}}(z_k)), \quad (15)$$

where L_{rem} is remaining path arclength. The name is heuristic: h is neither a probability nor a guarantee. A second multi-source Dijkstra field propagates these boundary values inward with an obstacle-proximity penalty $\eta(p)$,

$$D_g(p) = \min_{m \in \partial \mathcal{B}_p} [h(m) + d_\eta(p, m)], \quad (16)$$

so narrow, low-clearance channels remain reachable but are less attractive.

Candidate endpoints are ranked by $D_g(m) + d_r(m)$ and spatially suppressed to remove nearby duplicates. Each retained endpoint is traced back to the robot by descending d_r . A connector test rejects branches that are geometrically separated at their endpoints but not separated by an obstacle; at most three local branches are retained and tracked between field rebuilds. For branch P_m , a branch-following policy produces a horizon-length ancillary mean

$$\boldsymbol{\mu}_m = \mathcal{A}(P_m, x_0) = [\mathbf{u}_{m,0}, \dots, \mathbf{u}_{m,T-1}], \quad (17)$$

subject to velocity and acceleration limits. Its full rectangular-footprint rollout is collision-checked before it may seed a proposal group.

C. Grouped sampling and per-group free energy

Rather than running an independent optimiser per homotopy class, TG-MPPI partitions a single rollout batch into groups, one per active guidance mode plus a fallback group warm-started from the previous solution. Each group g is warm-started from its own guidance trajectory and scored by its own free energy

$$F_g = c_g^{\min} - \lambda \log \frac{1}{|g|} \sum_{i \in g} \exp\left(-\frac{1}{\lambda} \left(S^{(i)} - c_g^{\min}\right)\right), \quad (18)$$

and the group with the lowest F_g is selected, with a switch margin to suppress chattering.

A consequence of (18) deserves note: $c_g^{\min}$ is a minimum over $|g|$ samples, so its expectation decreases with group size roughly as $-\sigma\sqrt{2\ln|g|}$. A large fallback group therefore enjoys a systematic head start over small guidance groups, independent of trajectory quality. Both an unequal (*legacy*) and an equal-share allocation are evaluated in Section VII.

D. Space-time routes

Spatial pseudopods distinguish routes around static geometry but cannot encode that the same position may be blocked at one arrival time and free at another. TG-MPPI therefore constructs a coarser time-expanded graph whose vertices are (p, k) , with $\tau_k = k\Delta t_\ell$. Every transition advances one time layer and either moves to an eight-connected neighbour or waits in place. For constant-velocity obstacle prediction

$$\hat{o}_j(\tau_k) = o_j + v_j \min(\tau_k, T_{\text{pred}}), \quad (19)$$

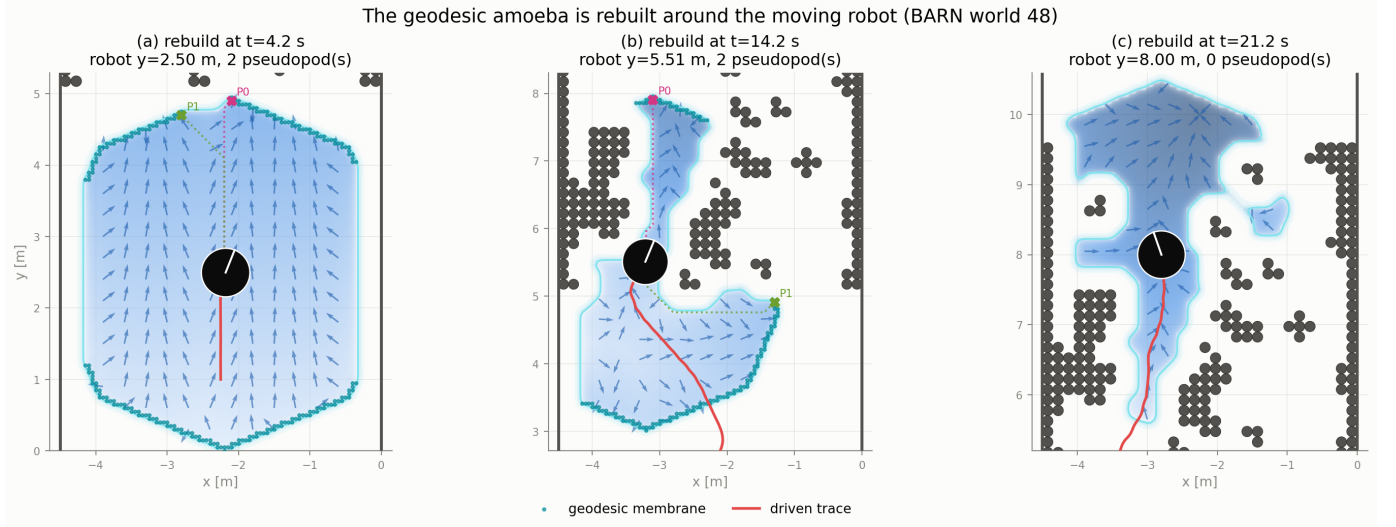


Fig. 1. Sandbox execution in BARN world 48. The geodesic body (blue), membrane (cyan), navigation directions (arrows), retained pseudopods (dotted curves) and driven trace (red) are shown at three field rebuilds. This is an implementation-derived explanatory visual, not a scored Nav2 benchmark trial.

a space–time vertex is free only when both the static footprint constraint and

$$\|p - \hat{o}_j(\tau_k)\|_2 > r_p + r_j \quad \forall j \quad (20)$$

hold. Diagonal moves additionally forbid corner cutting.

Dijkstra search is run twice with different costs for the wait transition. A low wait cost favours yielding until the crossing clears; a high wait cost favours continued motion and hence a spatial detour. This produces candidate routes π_{wait} and π_{detour} , which are converted to ancillary control means by the same constrained branch-following procedure as the spatial pseudopods. The search is proposal generation, not collision avoidance by itself: the predicted-obstacle critic still scores every MPPI rollout. Only routes that pass the winding-based distinctness test below are added as separate groups.

Figure 2 visualises the construction in both the ground plane and the time-expanded volume. A moving obstacle becomes a tube in (x, y, t) ; waiting and detouring pass different sides of that tube even when their two-dimensional projections partially overlap.

E. Homotopy distinctness

Two candidate routes are only worth optimising separately if they are homotopically distinct. Let $\mathbf{p}(t)$ be a route and $\mathbf{o}(t)$ the predicted position of an obstacle. The normalised winding number

$$\lambda_w = \frac{1}{2\pi} \sum_{k=1}^K \text{wrap}(\theta_k - \theta_{k-1}), \quad \theta_k = \text{atan2}(\mathbf{p}_k - \mathbf{o}_k), \quad (21)$$

accumulates the relative angle over the whole route, where $\text{wrap}(\cdot)$ maps to $(-\pi, \pi]$. A route is taken to have passed the obstacle when $|\lambda_w| \geq 1/(4\pi)$, and two routes are distinct when both pass it with opposite sign. This follows the winding-number test of T-MPC [7], which reports it as statistically equivalent to an H-signature [5] at lower cost.

Section VII-C reports why an earlier synchronised-time side test, which required both routes to be near the same obstacle at the same time step, was structurally unable to detect the distinction it existed to test.

V. IMPLEMENTATION

TG-MPPI is implemented as a `nav2_core::Controller` plugin for ROS 2 Humble, loadable in place of the stock `nav2_mppi_controller`. The baseline is the unmodified upstream controller; no change was made to it at any point, so baseline and treatment differ only in the plugin selected at launch.

The controller runs at 20 Hz with $N = 2000$ samples, $T = 56$ steps, and $\Delta t = 0.05$ s. Its critic chain contains the stock constraint, costmap, goal, goal-angle, path-align, path-follow, path-angle and prefer-forward terms, augmented by the predicted-obstacle critic. The flow critic is loaded but disabled in the reported configuration, isolating the effect of ancillary sampling. Rollout and the supported critic operations use a LibTorch/CUDA backend; unsupported operations retain a CPU path and the controller falls back to CPU if CUDA is unavailable. The control-period budget is 50 ms.

The Nav2 implementation uses the grouped V3 estimator: costs are normalised within each proposal group, group free energies are compared, and only the selected group’s weighted control update is applied. Dwell time, a free-energy switch margin, confirmation cycles and per-mode warm starts suppress mode chattering. It does *not* apply the explicit proposal-density correction $\log p(U) - \log q(U)$ of full importance-corrected Biased MPPI. Consequently, the thesis evaluates a topology-guided grouped MPPI implementation, while the importance-corrected estimator remains a sandbox formulation rather than a claim about the deployed Nav2 controller.

A. Development sandbox and transfer to Nav2

The algorithm was first developed in an independent NumPy/PyTorch sandbox. It provides deterministic BARN-

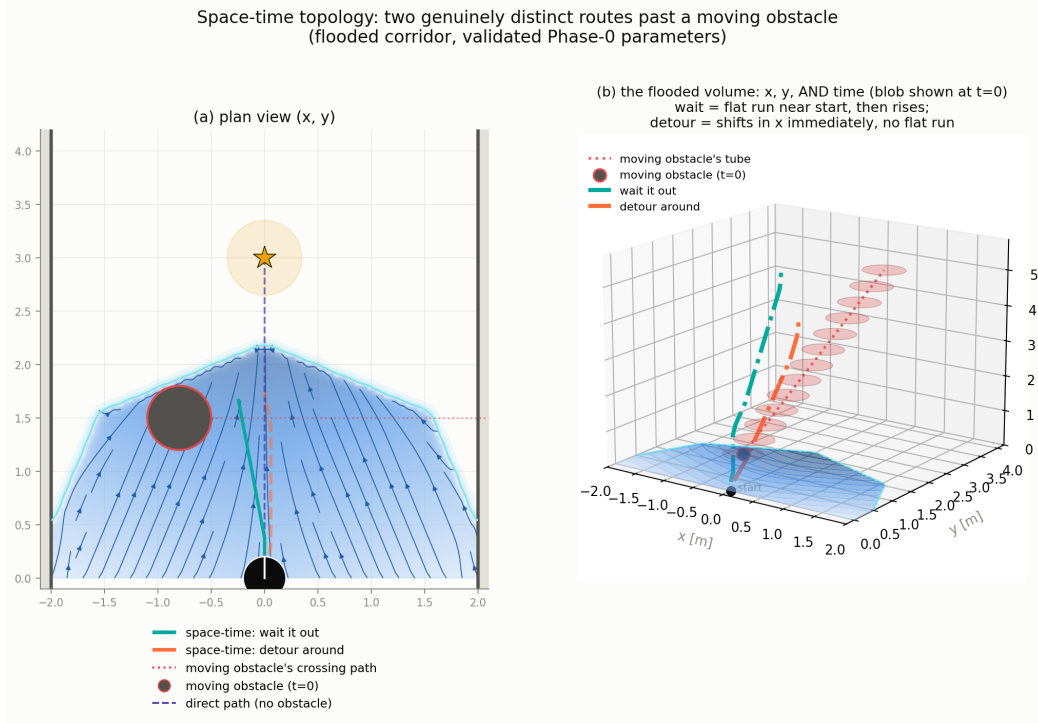


Fig. 2. Synthetic sandbox demonstration of space-time topology. Left: plan view of wait and detour candidates around a moving obstacle. Right: the same candidates in (x, y, t) , where the obstacle trajectory forms an exclusion tube. The diagram explains the proposal mechanism; quantitative claims are drawn only from the Gazebo benchmark.

map loading, ideal and analytical-slip motion models, A* references, geodesic-body and pseudopod extraction, grouped MPPI, dynamic-obstacle prediction, space-time search, visualisation and ablation harnesses. This environment was used to inspect algorithmic mechanisms and to reject unsafe or ineffective variants before integration.

The sandbox and Nav2 controller are not treated as interchangeable implementations. The final quantitative comparison uses the C++ `nav2_tgmppi_controller` plugin in Gazebo, with the same Nav2 lifecycle, costmaps, global plans and robot interfaces as the stock baseline. Sandbox results support design decisions and explanatory figures such as Fig. 1; they are not pooled with Gazebo trials in the headline statistics.

VI. EXPERIMENTAL SETUP

A. Robot and simulator

Experiments use the SLIP skid-steer model in Gazebo Classic 11 and ROS 2 Humble. Planning and contact scoring use its rectangular $0.84\text{ m} \times 0.68\text{ m}$ footprint. The benchmark launch sets $v_{x,\max} = 1.5\text{ m s}^{-1}$; the controller additionally enforces its configured acceleration and yaw-rate limits. Front and rear lidar data populate the local costmap, while localisation is supplied from Gazebo ground truth through the same map-odometry interface intended for motion-capture localisation on the physical platform. AMCL is therefore not part of the reported comparison.

B. Scenarios

Five scenarios (dyn1–dyn5) were generated over a common occupancy map, each with moving obstacles following polynomial tracks at scenario-dependent speeds. Fig. 3 shows the start pose, goal poses and obstacle tracks.

C. Conditions

The ablation isolates prediction from topology:

- A stock Nav2 MPPI (unmodified baseline);
- B TG-MPPI, full, legacy (unequal) sample split;
- B' TG-MPPI, full, equal sample split;
- D plain MPPI plus the predicted-obstacle critic, topology disabled.

Condition D is the critical control: it carries the obstacle prediction but none of the topological machinery, so the difference $B - D$ measures what topology contributes over prediction alone.

D. Protocol and metrics

Every trial starts from a fixed simulation time so that all conditions encounter identical obstacle phases; goals are issued in a fixed sequence with no preemption. Contacts are scored from ground truth (robot footprint against obstacle discs) rather than from the costmap, so a condition cannot score well by failing to perceive an obstacle. Each condition is run 5 scenarios $\times$ 3 repetitions = 15 trials, four goal legs each. Continuous per-trial metrics are compared with two-sided Mann–Whitney U tests and binary outcome counts with

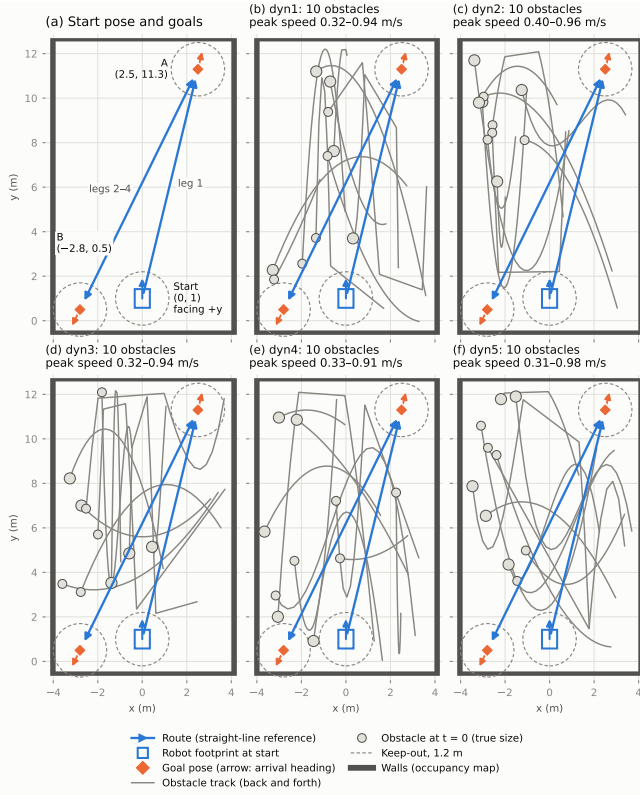


Fig. 3. Benchmark start pose, goal poses and obstacle tracks for dyn1–dyn5 over the occupancy map used in all trials.

two-sided Fisher exact tests. Statistical significance is assessed at $\alpha = 0.05$ without treating repeated controller cycles as independent observations. In addition to success, the report records effective speed, driven-path ratio, minimum clearance, first-contact cause, recovery count, controller cycle time and deadline misses. Unattempted legs after early trial termination count as failures in planned-leg success rates.

E. Physical SLIP validation plan

Following the simulation benchmark, a limited matched validation is planned on the physical SLIP platform. Localisation will be supplied by Vicon and obstacle observations by the onboard lidar pipeline rather than Gazebo ground truth. The objective is transfer validation—whether the controller runs at its required rate and preserves the simulation safety trend under real skid, latency and sensing—not retraining or retuning on the test course.

Trials will progress from an empty-path check to a static narrow passage and then to a controlled moving-obstacle crossing. Stock Nav2 MPPI (A), prediction only (D), and the selected TG-MPPI configuration (B or B') should receive the same start, goal, speed limit and obstacle script, with controller order counterbalanced across repetitions. Recorded ROS bags should include Vicon pose, raw and filtered scans, odometry, velocity commands, global and local plans, costmaps, tracked-obstacle states, controller diagnostics and TF. The primary hardware outcomes are goal completion without intervention, minimum Vicon-based clearance, traversal time, path length,

emergency stops and control-cycle latency. A human-operated emergency stop, conservative initial speed and a non-human soft obstacle are required for the moving trial.

VII. RESULTS

A. Prediction accounts for the gain over the baseline

Table I shows a large and unambiguous separation between the stock baseline and every condition carrying obstacle prediction. The baseline completed 19 of 45 legs and made contact on 32; all three prediction-carrying conditions completed 53–57 of their legs with contact on 9–10. **[TODO: insert the Mann–Whitney/Fisher statistics and p -values.]**

B. Topology relative to prediction

In the original run, conditions B, B' and D were statistically indistinguishable on every headline metric at $n = 15$, while the topological machinery added approximately 4 ms per control cycle (19.2 vs 15.3 ms). That run used the invalid synchronised-time distinctness test, so it cannot provide the final verdict on the implemented space–time topology layer. **[PENDING: Replace this paragraph with the B–D comparison from the corrected winding-number run. State whether the result is a statistically supported effect, a bounded null result, or an underpowered comparison; report an effect size and confidence interval rather than relying on p alone.]**

C. Space-time distinctness

[PENDING: these numbers are from the synchronised-time side test and are superseded by the winding-number re-run; regenerate this subsection from thesis_dyn_winding before submission.] Across 25 000 control cycles of conditions B and B', the space-time search found a feasible route on 86–90 % of cycles, but 96.7 % of those routes were classified non-distinct from a route the pseudopod stage had already produced, at a cost of 5.1 ms to 5.3 ms per cycle (27–28 % of the control budget).

That measurement, however, reflects the distinctness test, not the routes. The synchronised-time side test required both routes to lie within a relevance radius of the same obstacle at the same time index. A pass-before route and a pass-behind route are near the obstacle at *different* time indices by construction, so the test was left unevaluated at almost every layer and defaulted to “not distinct”. It was structurally blind to precisely the distinction it existed to detect. Replacing it with the winding number of (21), which accumulates each route's relative angle independently over its whole length, removes the synchronisation requirement. **[TODO: report the corrected distinctness rate and whether B vs D changes.]**

VIII. DISCUSSION AND LIMITATIONS

The ablation is designed to prevent a useful prediction mechanism from being mistaken for evidence of topological reasoning: condition D receives the same predicted-obstacle critic as TG-MPPI but no pseudopod or space–time proposal. Whether topology adds measurable value must therefore be

TABLE I
DYNAMIC BENCHMARK, 60 TRIALS (15 PER CONDITION, FOUR LEGS EACH). EFFECTIVE SPEED IS STRAIGHT-LINE DISTANCE OVER TIME ON SUCCESSFUL LEGS. CONTACTS ARE SCORED FROM GROUND TRUTH.

	Description	Legs ok	Eff. speed (m s ⁻¹)	Legs w/ contact	Min clearance (m)	Recoveries/leg	Cycle (ms)
A	Stock Nav2 MPPI	19/45	0.76 ± 0.28	32/45	-0.038 ± 0.403	12.00 ± 8.56	—
B	TG-MPPI, legacy split	57/59	0.72 ± 0.16	10/59	0.274 ± 0.280	0.51 ± 1.83	19.2 ± 0.5
B'	TG-MPPI, equal split	57/59	0.76 ± 0.17	9/59	0.287 ± 0.242	0.97 ± 3.76	19.0 ± 0.3
D	Plain MPPI + prediction	53/57	0.71 ± 0.20	9/57	0.276 ± 0.291	1.75 ± 5.12	15.3 ± 0.2

inferred from B versus D in the corrected winding-test run, not from TG-MPPI versus the stock baseline. A non-significant B–D difference would be a null result for this benchmark and sample size, not proof that topology can never help. Resolving a small effect would require more repetitions and scenes deliberately containing two feasible classes with different temporal consequences.

Scenario coverage is limited to five scripted obstacle patterns over one map. Although restarting Gazebo and fixing the first-goal simulation time makes conditions comparable, it does not establish generality across building layouts, crowd densities or obstacle behaviours. In particular, the obstacles follow known polynomial tracks, whereas the controller observes position and velocity and extrapolates constant velocity. Abrupt acceleration or reversal therefore produces prediction error. The reported trials also use perfect ground-truth obstacle observations without perception noise, delay, missed detections or identity switches. The available tracking-quality perturbations and a real lidar tracker remain validation work rather than evidence included in the main result.

The main controlled experiment is simulation-only. Gazebo contact and skid behaviour cannot substitute for hardware tests of wheel slip, actuator delay and safety stops; the planned physical trials are therefore reported as a separate transfer study and must not be described as completed until their bags have been scored. Furthermore, the winding classifier is a pragmatic finite-route test rather than a complete topological invariant for every local geometry. A route that passes far from an obstacle can accumulate less than the $1/(4\pi)$ pass threshold and be classified as not having passed it. This failure mode should be reported alongside the corrected distinctness rate.

IX. CONCLUSION

This thesis presented TG-MPPI, a Nav2 controller that turns a robot-centred geodesic flood field into collision-validated ancillary MPPI proposals and can augment them with time-expanded wait/detour routes. A reproducible Gazebo benchmark compares it with an untouched stock Nav2 MPPI controller and, through a prediction-only condition, separates obstacle prediction from topology. **[PENDING: Replace with the final quantitative conclusion from thesis_dyn_winding: state the stock-baseline gain, then state whether B separated from D under the corrected winding classifier.]** The result should be interpreted within the limits of perfect obstacle tracking, five simulated scenarios and no physical-robot validation.

ACKNOWLEDGMENT

[TODO: supervisor, lab, compute.]

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